SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)

DepartmentofComputerScienceandEngineering

ASSESSMENTTOOL2–TraversalChallenge

CourseCode:	CSA0305	**CourseTitle:**	DataStructures
COAssessed:	CO3–Precision(BL3,BL5)	**Assessment:**	AssessmentTool2–Traversal Challenge
Weightage:	30%of CIA	**Total Marks:**	100
CO3Statement:	*SolveproblemsforoptimizeddatastorageandretrievalusingBinarySearchTrees,AVLTrees,B-Trees, Red-Black Trees, and Splay Trees.*		
StudentName:	M. VAISHNAV	**Reg. No.:**	192524251
Section/Batch:	AIandDS/2025	**Date:**	03/08/2026

CodeofConduct

I M.VAISHNAV (192524251) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

SignatureoftheStudent: *M.VAISHNAV (192524251)*

Instructions

- Thisassessmenttoolcontains5TraversalChallengequestions,Total:100Marks.
- Studentscanuseanyonline/offlineTraversalChallenge.Demonstratetree.
- Whenastudentanswerusingsimulator,inevaluationtheanswershouldhavecapturedscreenshots after every major operation
- SubmitthereportinPDFformatwithproperlylabelledscreenshotsasproofofcompletion.

Section/Topic	Focus	QNos	Marks
Tree Traversals	Traversaltypes,Inorder,PostorderandPreorder	5	100
	GRAND TOTAL:	**100Marks**	

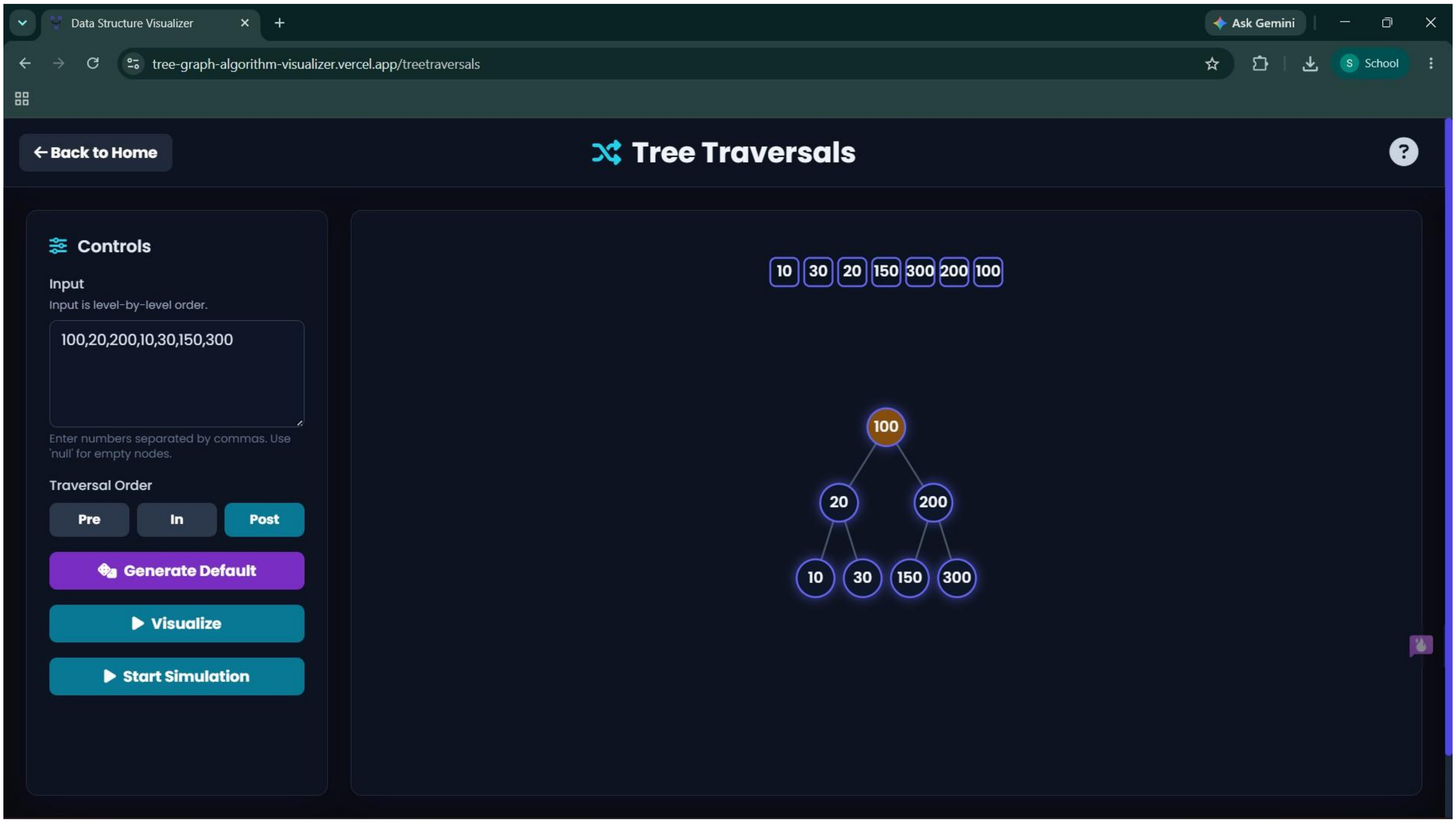
TraversalChallenge

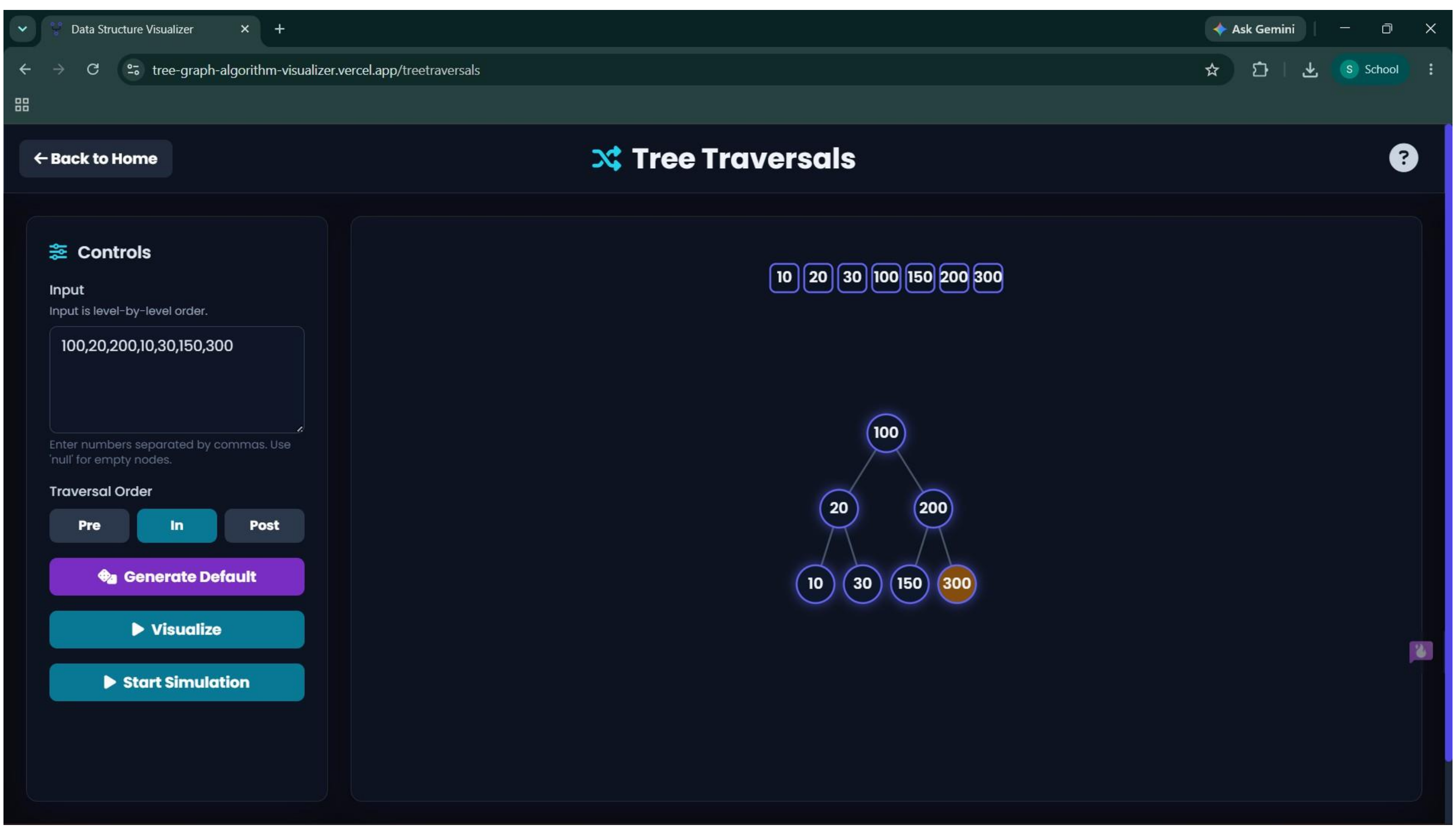
Rubricsfor Calculation

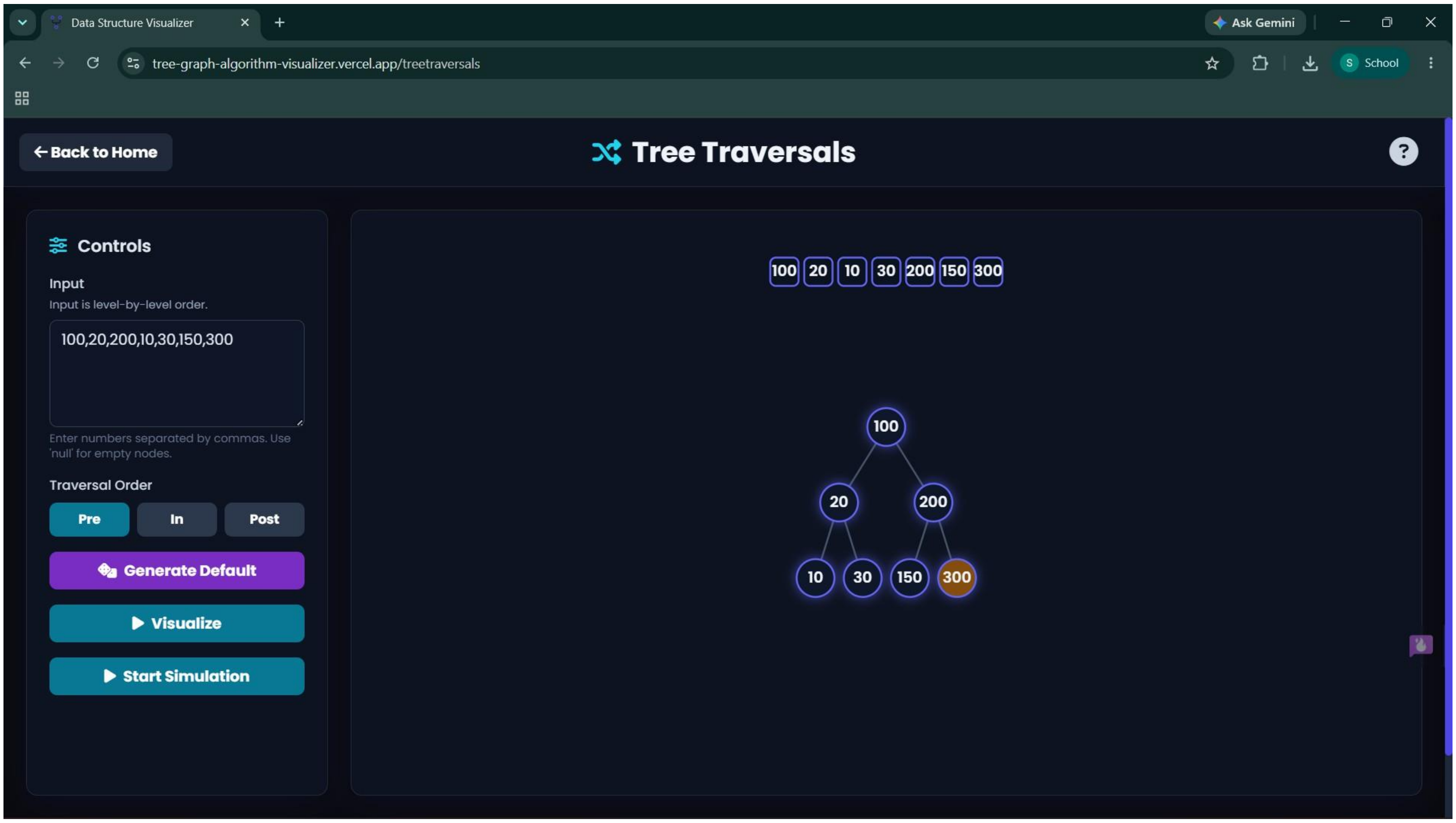
Criteria	Weightage	Level4–Exemplary	Level3–Proficient	Level 2 – Developing	Level 1 – Beginning
ConceptualUnderstanding	20	Demonstratesthoroughunderstanding of all coreDataStructuresconcepts;answersaccuratelyreflectdeepknowledge	Showsgoodunderstandingwithminorconceptualgaps	Basicunderstanding;severalmisconceptionspresent	Limitedunderstanding;majorconcepts misunderstood
TraversalAccuracy	20	Correctlycompletesalltreetraversals(Preorder,Inorder, Postorder, LevelOrder) without errors.	Completesmosttraversalscorrectly withoneminorerror.	Correctlyperformsonlysometraversalswithmultipleerrors.	Unabletoperformmosttraversalscorrectly.
Selectionof Appropriate Traversal	30	Correctly selects theappropriatetraversalforevery given problemscenario.	Selectsthecorrect traversalformost scenarios.	Selectsappropriate traversal foronly afewscenarios.	Frequentlyselectsincorrecttraversal methods.
Completion&TimeEfficiency	20	Completes all challengeswithin the allotted timewhile maintainingaccuracy.	Completesmostchallengeswithinthe time limit.	Completesonlypartof the challenge orexceeds the timelimitslightly.	Unabletocompletethechallengewithin the allottedtime.
EvidenceSubmission	10	Uploadscompletescreenshotsshowingsuccessfulcompletion,score,andstudentdetails.	Uploadsscreenshotswithminormissinginformation.	Uploadsincompleteorunclearscreenshots.	Novalidevidencesubmitted.

FacultyIn-charge	CourseCoordinator	ProgramDirector
Signature: ______________	Signature: ______________	Signature: ______________
Date: _________________	Date: _________________	Date: _________________

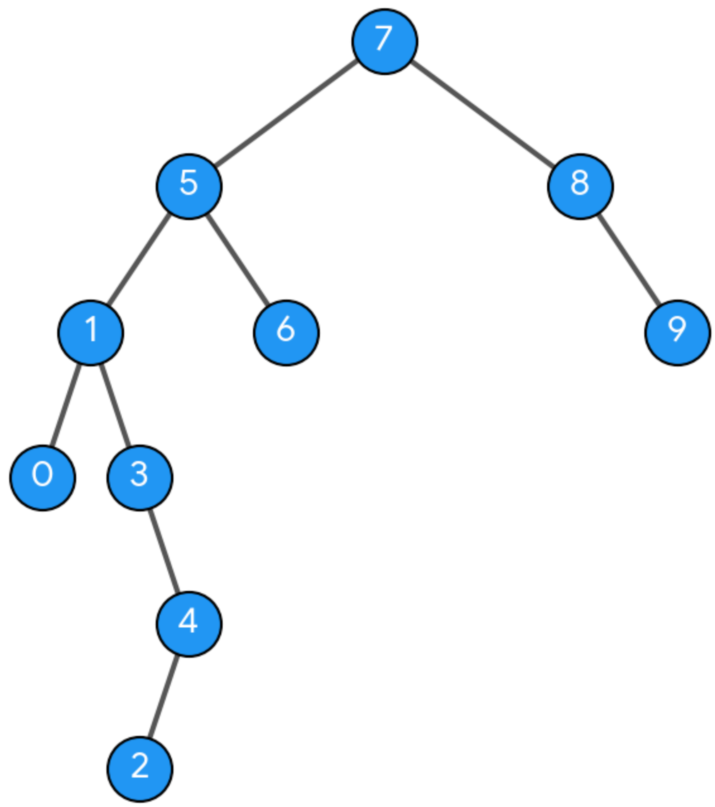
1. Writethetraversalsequencesforfollowingtrees



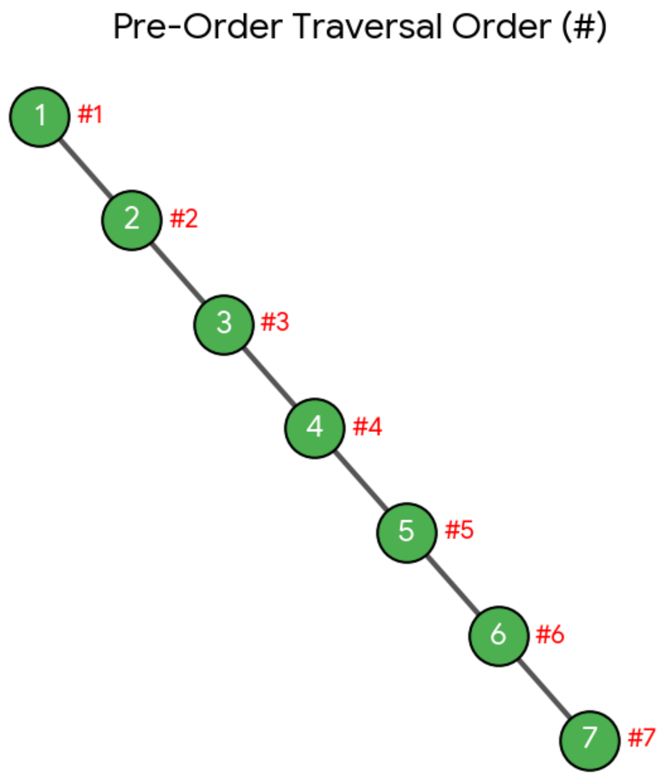


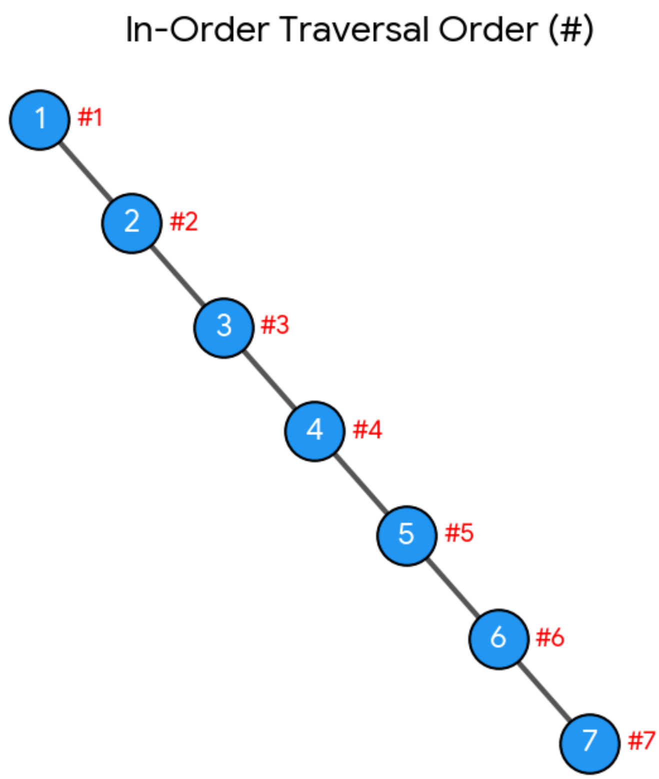

Data Structure Visualizer
Ask Gemini
tree-graph-algorithm-visualizer.vercel.app/treetraversals
School
Back to Home
Tree Traversals
Controls
Input
Input is level-by-level order.
100,20,200,10,30,150,300
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
100 20 10 30 200 150 300
100
20
200
10
30
150
300

2. Assumethedigits7,5,1,8,3,6,0,9,4,2areaddedinthatsequenceintoabinarysearchtreethatisinitially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?

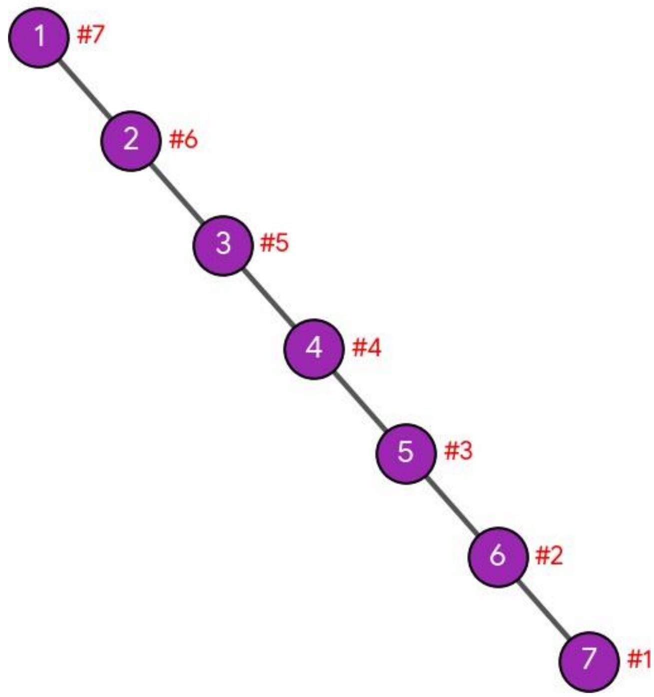


3. WritetheInorder,Preorder&Postordertraversalsequencesforfollowingtrees



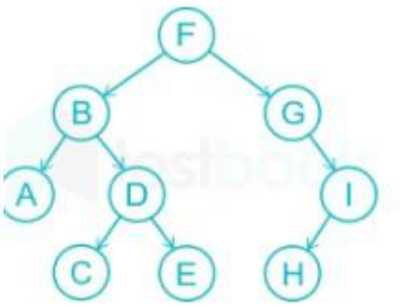


Post-Order Traversal Order (#)



Question No: 4

What is the sequence of nodes when applying in-order traversal on the binary tree given below?



1. A, B, C, D, E, F, I, H, G

2. A, C, D, E, B, F, G, H, I

3. A, B, C, D, E, F, G, H, I

4. I, H, G, F, E, D, C, B, A

Thecorrectoptionis**3.A,B,C,D,E,F,G,H,I**. **Step-by-Step Breakdown**

In-ordertraversalfollowstherule:**LeftSubtree\(\rightarrow\)Root\(\rightarrow\)RightSubtree**.

1. **LeftSubtreeof F (rootedat B):**
   - Goto theleftmost node:**A**
   - Visit root: **B**
   - Traverserightsubtreeof B(rootedatD):
     - Leftchild:**C**
     - Root:**D**
     - Rightchild:**E**
   - *Sequenceso far:* A, B, C, D, E
2. **VisittheMainRoot:F**
   - *Sequenceso far:* A, B,C, D, E, F
3. **RightSubtreeofF(rootedatG):**
   - LeftchildofG:None
   - Visit root: **G**
   - Traverserightsubtreeof G(rootedatI):
     - Leftchild:**H**
     - Root:**I**
   - *Sequenceso far:*G, H,I

Final Sequence

Combiningallstepsgives:**A,B, C,D,E,F,G,H, I** answerthequestion

Thecorrectoptionis**3.A,B,C,D,E,F,G,H,I**. **Step-by-Step Breakdown**

In-ordertraversalfollowstherule:**LeftSubtree\(\rightarrow\)Root\(\rightarrow\)RightSubtree**.

1. **LeftSubtreeofF(rootedatB):**
   - Gototheleftmostnode:**A**
   - Visitroot:**B**
   - TraverserightsubtreeofB(rootedatD):
     - Leftchild:**C**
     - Root:**D**
     - Rightchild:**E**
   - *Sequencesofar:*A,B,C,D,E
2. **VisittheMainRoot:F**
   - *Sequencesofar:*A,B,C,D,E,F
3. **RightSubtreeofF(rootedatG):**
   - LeftchildofG:None
   - Visitroot:**G**
   - TraverserightsubtreeofG(rootedatI):
     - Leftchild:**H**
     - Root:**I**
   - *Sequencesofar:*G,H,I

Final Sequence

Combiningallstepsgives:**A,B,C,D,E,F,G,H,I**

Ifyou'dlike,Icanalsoshowyouhowthe **pre-order**or**post-order**traversalswouldlookforthissametree.

Assumethedigits7,5,1,8,3,6,0,9,4,2areaddedinthatsequenceintoabinarysearchtreethatis initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence generate the image for this question

TofindthetraversalsWefirstconstructtheBinarySearchTree(BST)byinsertingthenumbers7,5,1,8,3,6, 0, 9, 4, 2 one by one using standard ordering (smaller values to the left, larger values to the right):

- **7**: Root
- **5**:Leftof7
- **1**:Leftof5
- **8**:Rightof 7
- **3**:Rightof 1
- **6**:Rightof 5
- **0**:Leftof1
- **9**:Rightof 8
- **4**:Rightof 3
- **2**:Leftof4

HereisthevisualstructureoftheresultingBST:

TraversalSequences

Dependingonthetraversalmethodrequired,thesequencesare:

- **In-Order**(Left\(\rightarrow\)Root\(\rightarrow\)Right):**0,1,2,3,4,5,6,7,8,9***(Alwayssortedforany valid BST)*
- **Pre-Order**(Root \(\rightarrow\) Left\(\rightarrow \)Right):**7,5, 1,0, 3,4, 2,6, 8, 9**
- **Post-Order**(Left\(\rightarrow \)Right \(\rightarrow\) Root):**0, 2,4, 3,1, 6,5, 9,8, 7**

Ifyouwant,letmeknowifyouwouldlikemetolistoutthe**step-by-steptrace**foranyspecificoneof these traversals!

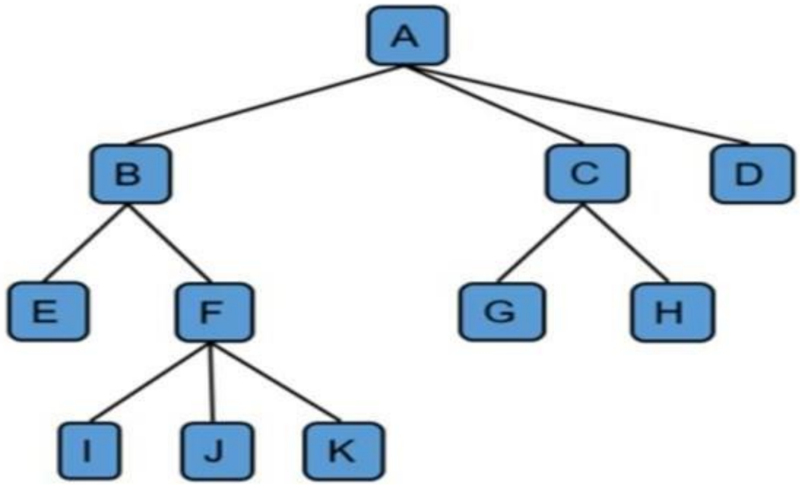
writeinorder,preorder,postorderfor1,2,3,4,5,6,7 generatetheimages seperately

Insertingthenumbers1,2,3,4,5,6,7intoaninitiallyemptyBinarySearchTreesequentiallycreatesa **right-skewed (degenerate) tree**, where every node only has a right child.

1.Pre-OrderTraversal(Root\(\rightarrow\)Left\(\rightarrow\)Right)

Sequence:1,2, 3,4,5, 6,7

Question No: 5



1. **1.TraverseLeftmostSubtreeof rootA(rootedatB):**
    - LeftmostchildofBis**E**.
    - Visittheparentroot:**B**.
    - TraverseremainingchildsubtreesofB(rootedatF):
        - ProcessleftmostchildofF:**I**.
        - Visittheparentnode: **F**.
        - Processnextchildrenfromlefttoright:**J**,then**K**.
    - *Resultingsegment:*E,B,I, F,J,K
2. **2.VisitMain RootNode: A**
    - *Combinedsegment:*E,B,I, F, J,K,A
3. **3.TraverseRemainingSubtreesofA(rootedat Cand D):**
    - **SubtreeunderC:**
        - Processleftmostchildof C:**G**.
        - Visitparentnode:**C**.
        - ProcessrightchildofC: **H**.
    - **NodeD:**Hasnochildren,simply visit**D**.
    - *Resultingsegment:*G,C,H,D

FinalCombinedSequence
E, B, I, F, J, K, A, G, C,H, D

1. Pre-OrderTraversal(Root\(\rightarrow\)Leftto RightSubtrees)
Rule: Visittheparentnodefirst, thenmovethrough itschildsubtreesfromleftto right.

1. **Visitmainroot:**A
2. **MovetoA'sfirstchild(B)anditssubtree:**
    - Visit B
    - VisitB'sfirstchild: E
    - VisitB'ssecondchild(F) andits subtree:
        - Visit F
        - VisitF'schildrenfromlefttoright:I,J,K
    - *SubtreeBpathcompleted:*B,E,F, I,J,K
3. **MovetoA'ssecondchild(C)andits subtree:**
    - Visit C
    - VisitC'schildrenfromlefttoright:G, H
    - *SubtreeCpathcompleted:*C,G,H

4. **MovetoA'sthirdchild (D):**
    - Visit D

    CompletePre-OrderSequence:
    A, B, E, F, I, J, K, C, G,H, D

2. Post-OrderTraversal(LefttoRightSubtrees \(\rightarrow\) Root)
Rule: Completelytraverseallchildsubtreesfromlefttorightbeforevisitingtheirparentnode.

1. **ExploreA'sfirstchild(B)anditssubtreecompletely first:**
    - GodowntoB'sfirstchild:E(hasnochildren,sorecordE)
    - GotoB'ssecondchild(F)andclearitschildrenfirst: I,J,K
    - Recordparentofthosechildren:F
    - NowthatallofB'schildrenarecleared,recordparent:B
    - *SubtreeBpathcompleted:*E, I,J,K,F,B
2. **ExploreA'ssecondchild(C)anditssubtreecompletely:**
    - ClearC'schildrenfromlefttoright:G, H
    - Recordparent:C
    - *SubtreeCpathcompleted:*G,H,C
3. **ExploreA'sthirdchild(D):**
    - Hasnochildren, sorecord D
4. **Finally,recordthemainrootnode:**
    - Visit A

    CompletePost-OrderSequence: E,
    I, J, K, F, B, G, H, C, D, A